

A Bidefect Counting Reduction for Aubrun’s Partial-Transpose Moment Bound

Kieran McShane

June 7, 2026

Abstract

We isolate a pure combinatorial reduction underlying Aubrun’s moment method for partial transposes of random induced states. The reduction replaces a single total-defect count by a two-parameter count indexed by the two oriented Cayley defects of a Wick permutation. This bidefect form is the one compatible with the shifted scalar bases appearing in the analytic estimate.

The main result of this draft is conditional and deliberately explicit: the bidefect envelope follows from two large-gap Biane adjacent-link inputs, one nested predecessor-skip collision input, and one Chapuy-type genus recurrence. The finite gap-three Biane branches and the direct predecessor-skip branch are separated from these four inputs and accounted for by elementary incidence-tree reductions. A new branch-normalized lemma records the canonical first/last return interval in every genuinely nested predecessor skip, and the current reduction exposes the nested collision input through this return interval. The remaining four inputs are stated in mathematical form rather than hidden in notation.

Contents

1 Purpose and Scope	1
2 Permutations, Defects, and the Bidefect Envelope	2
3 Why the Count Must be Bidefect	3
4 The One-Sided Genus Route	4
5 The Biane Contour Mechanism	4
6 Finite Gap-Three Biane Reductions	6
7 The Conditional Bidefect Theorem	6
8 What Remains for an Unconditional Article	8
A Synchronization with the Formal Development	8

1 Purpose and Scope

Aubrun’s proof of the partial-transpose moment estimate is a moment-counting argument over Wick permutations. The analytic side assigns to each permutation π a power of the matrix dimension and the environment dimension; the combinatorial side must show that the number of permutations with a given loss of rank is not too large. A tempting formulation counts only

the total loss. That formulation is not stable enough for the final scalar estimate, because the two losses are paid by two different shifted bases.

This article develops the corrected two-parameter formulation. It is written as an ordinary mathematical draft. The formalization status is recorded only in Appendix A; the main body is meant to be read without Lean.

Remark 1.1 (Honesty of the present version). The unconditional U-AUB-02 theorem is not closed in the formal development at the time of this draft. The theorem below is therefore a reduction theorem: it says exactly which four combinatorial inputs remain. It is nevertheless a usable article skeleton, because those inputs are now localized and no longer confused with rank-budget bookkeeping or finite checked branches.

2 Permutations, Defects, and the Bidefect Envelope

Fix $m \geq 0$, put $n = m + 1$, and let

$$\gamma = (0\,1\,\dots\,m) \in \mathfrak{S}_n$$

be the long cycle. For a permutation σ , write $\text{cyc}(\sigma)$ for the number of cycles, including fixed points. The Cayley length is

$$|\sigma| = n - \text{cyc}(\sigma).$$

In the partial-transpose Wick expansion, the surviving index identifications can be encoded by a permutation $\pi \in \mathfrak{S}_n$. This is the only meaning of “Wick permutation” in this article. The sample index contributes $2\,\text{cyc}(\pi)$ in the balanced $s \sim \lambda d^2$ regime, while the two matrix-index directions contribute $\text{cyc}(\gamma\pi)$ and $\text{cyc}(\pi\gamma^{-1})$. This is the source of the weighted rank in Section 3.

Definition 2.1 (Oriented defects). For $\pi \in \mathfrak{S}_n$, define the two oriented defects

$$\delta_+(\pi) = n + 1 - \text{cyc}(\pi) - \text{cyc}(\gamma\pi), \quad \delta_-(\pi) = n + 1 - \text{cyc}(\pi) - \text{cyc}(\pi\gamma^{-1}).$$

The bidefect class and its cardinality are

$$\mathcal{B}_m(a, b) = \{\pi \in \mathfrak{S}_n : \delta_+(\pi) = a, \delta_-(\pi) = b\}, \quad N_m(a, b) = |\mathcal{B}_m(a, b)|.$$

The inequalities

$$\text{cyc}(\pi) + \text{cyc}(\gamma\pi) \leq n + 1, \quad \text{cyc}(\pi) + \text{cyc}(\pi\gamma^{-1}) \leq n + 1$$

are just the Cayley triangle inequality written in cycle-count form. Hence δ_+ and δ_- are non-negative integers.

Definition 2.2 (Aubrun bidefect envelope). Let $P(Q)$ be the polynomial scalar envelope used in the Wick estimate. In the formal development it is denoted $P(Q)$ with

$$Q = 2(m + 1).$$

Define

$$\mathcal{E}_m(a, b) = 2^{2(m+1)} \binom{m+3}{a} \binom{m+1}{b} P(2(m+1))^{a+b}.$$

The bidefect count target is

$$N_m(a, b) \leq \mathcal{E}_m(a, b)$$

for

$$0 \leq a \leq m + 3, \quad 0 \leq b \leq m + 1.$$

Remark 2.3. The ranges are part of the theorem. One can state an all- (a, b) version only after proving that the out-of-range classes are empty or harmless. The current reduction deliberately avoids that extra claim.

3 Why the Count Must be Bidefect

The Wick weight contains three cycle-count contributions: one from the sample index and two from the left and right matrix indices. In the balanced regime the weighted rank has the form

$$R(\pi) = 2 \operatorname{cyc}(\pi) + \operatorname{cyc}(\gamma\pi) + \operatorname{cyc}(\pi\gamma^{-1}).$$

The rank budget is

$$R(\pi) + \delta_+(\pi) + \delta_-(\pi) = 2n + 2.$$

Equivalently, if the total defect is

$$\Delta(\pi) = 2n + 2 - R(\pi),$$

then

$$\Delta(\pi) = \delta_+(\pi) + \delta_-(\pi).$$

The identity explains both the success and the failure of the old route. A single-defect count sees the correct rank loss, but it forgets which side of the scalar estimate pays that loss. The bidefect envelope keeps the two losses separate until the final scalar absorption.

Proposition 3.1 (Shifted-base absorption, interface form). *Assume the bidefect count target*

$$N_m(a, b) \leq \mathcal{E}_m(a, b)$$

for all a, b in the stated ranges. Then the corresponding double sum over Wick permutations is bounded by the shifted scalar bases used in Aubrun's moment estimate, rather than by a common maximum base.

Proof. After grouping permutations by (a, b) , the count contributes the two binomial factors

$$\binom{m+3}{a}, \quad \binom{m+1}{b}.$$

The analytic summand has the schematic form

$$A^{m+3-a} B^{m+1-b} Q^{a+b},$$

where A and B are the two scalar bases associated with the two matrix directions and $Q = P(2(m+1))$ is the polynomial loss budget. Thus the a -sum is bounded by

$$\sum_a \binom{m+3}{a} A^{m+3-a} Q^a = (A + Q)^{m+3},$$

and the b -sum is bounded by

$$\sum_b \binom{m+1}{b} B^{m+1-b} Q^b = (B + Q)^{m+1}.$$

The important point is that the exponent a is absorbed on the left side and b on the right side; replacing them by $a + b$ before the scalar step loses the shifted-base structure needed for the final estimate. $\square$

4 The One-Sided Genus Route

The bidefect estimate is supplied from the plus side. A bidefect class embeds in its plus-defect class:

$$\mathcal{B}_m(a, b) \subseteq \{\pi : \delta_+(\pi) = a\}.$$

Thus the bidefect problem is reduced to a one-sided count for δ_+ , with the missing b -dependence restored by the binomial scalar envelope.

Definition 4.1 (Left-tight permutations). A permutation $\pi \in \mathfrak{S}_n$ is left-tight if

$$\text{cyc}(\pi) + \text{cyc}(\gamma\pi) = n + 1.$$

Equivalently, $\delta_+(\pi) = 0$.

Proposition 4.2 (Genus-zero base). *The zero plus-defect class is the class of left-tight permutations. Therefore the genus-zero count is bounded once left-tight permutations inject into noncrossing partitions:*

$$\#\{\pi : \delta_+(\pi) = 0\} \leq |\text{NC}(n)| \leq 4^n.$$

Proof. The first statement is the definition of δ_+ . The second follows if the cycle partition of a left-tight permutation is noncrossing and determines the permutation. The number of noncrossing partitions on n points is the n -th Catalan number, and the elementary bound $C_n \leq 4^n$ gives the displayed estimate. $\square$

For positive plus-defect the intended interpretation is via unicellular maps:

$$\delta_+(\pi) = 2g(\pi).$$

The map model is the standard two-permutation encoding of a one-face map: the long cycle records the face order and the permutation π records the vertex identifications. Euler's formula then converts the cycle-count defect into twice the genus. The present article does not prove the full map-model equivalence; it isolates it as part of the Chapuy input below. The target recurrence is a Chapuy slicing estimate of the form

$$N_{m,g} \leq (2n)^3 N_{m,g-1}, \quad n = m + 1,$$

where

$$N_{m,g} = \#\{\pi : \delta_+(\pi) = 2g\}.$$

Together with the genus-zero base, this gives

$$N_{m,g} \leq 4^n (2n)^{3g}.$$

This is the enumerative heart of the remaining U-AUB-02 proof.

5 The Biane Contour Mechanism

The genus-zero injection depends on recovering a left-tight permutation from its cycle partition. This recovery is obstructed by predecessor skips. The proof route uses Biane's noncrossing geometry: from a skip one extracts a long interval whose contour has a repeated class; path uniqueness in the associated incidence tree then forces a contradiction.

Let

$$S_\pi = \text{cycle partition of } \gamma\pi, \quad T_\pi = \text{cycle partition of } \pi.$$

The incidence graph has left vertices the S_π -classes, right vertices the T_π -classes, and an edge for each point of $\{0, \dots, m\}$. In the left-tight noncrossing case this incidence graph is a tree. Hence two simple paths with the same endpoints must coincide.

Definition 5.1 (Contour over an interval). For an interval $[a, c] \subseteq \{0, \dots, m\}$, the Biane contour is the alternating path in the incidence graph obtained by following the cyclic successor steps

$$a \rightarrow a + 1 \rightarrow \dots \rightarrow c.$$

At each step the path crosses from an S_π -class to a T_π -class and back. A repeated contour class means that two distinct points of the interval give the same left vertex S_π , or two distinct interior points give the same right vertex T_π . In the incidence tree, such a repeat is the combinatorial event used to manufacture a second simple path.

Definition 5.2 (Predecessor skip). A non-fixed point x is a predecessor skip if the cyclic predecessor relation expected from the ordered cycle partition fails at x . Concretely there are two shapes.

- (i) Nonwrapping: $\pi x < x$, and the π -cycle of x contains a point strictly between πx and x .
- (ii) Wrapping: $\pi x \not< x$, and the π -cycle of x contains a point strictly below x or strictly above πx .

The direct-hit case is already elementary: if x and $\gamma(\pi x)$ lie in the same π -cycle, then one immediately obtains two distinct points lying simultaneously in one T_π -class and one S_π -class. This contradicts the tree transversality of a left-tight noncrossing permutation.

The remaining case is the genuinely nested case:

$$x \not\prec_\pi \gamma(\pi x).$$

The following lemma records the normalized interval data now available in all three skip shapes.

Lemma 5.3 (Branch-normalized nested return interval). *Let π be left-tight and let x be a non-fixed predecessor skip. Assume the direct-hit case is excluded:*

$$x \not\prec_\pi \gamma(\pi x).$$

Then there exist $a < c$ such that

$$c - a > 1, \quad a \sim_\pi c,$$

and the Biane contour over $[a, c]$ has a repeated S_π -class or a repeated T_π -class. Moreover the interval is canonical in its branch:

- (i) *in the nonwrapping branch, $a = \pi x$, and c is the first same- π return after πx ;*
- (ii) *in the wrap-below branch, $c = \pi x$, and a is the last same- π return below x ;*
- (iii) *in the wrap-above branch, $a = x$, and c is the first same- π return above πx .*

Proof. In the nonwrapping branch, take the least same-cycle point c satisfying

$$\pi x < c < x.$$

If $c = \pi x + 1$, then $c = \gamma(\pi x)$, contradicting the excluded direct-hit case. Hence $c - \pi x > 1$. Left-tightness then applies to the interval $[\pi x, c]$ and gives the repeated-contour obstruction.

In the wrap-below branch, nonfixedness and $\pi x \not< x$ imply

$$x < \pi x.$$

Take the largest same-cycle point $a < x$. Then $\pi x - a > 1$, and the interval $[a, \pi x]$ carries the same repeated-contour obstruction.

In the wrap-above branch, take the least same-cycle point $c > \pi x$. As in the nonwrapping case, the direct-hit exclusion rules out $c = \pi x + 1$. Since $x < \pi x < c$, the interval $[x, c]$ has length > 1 , and left-tightness again supplies the repeated-contour obstruction. $\square$

6 Finite Gap-Three Biane Reductions

The adjacent-link theorem is the local Biane statement used in the count. In one orientation, it has the following form. Suppose

$$a < b < c < d, \quad a \sim_{\pi} c, \quad b \sim_{\pi} d, \quad a \not\sim_{\pi} b.$$

The desired left adjacent link is

$$a \sim_{S_{\pi}} b.$$

The gap-two case is already settled. The gap-three cases split into small finite contour branches. The current reduction discharges these finite branches by incidence-tree uniqueness.

The branch accounting used in the reduction is as follows. Here $S(u, v)$ means $u \sim_{S_{\pi}} v$, and $T(u, v)$ means $u \sim_{\pi} v$.

Slice	Branches exposed by the contour repeat	Status in the reduction
$c - a = 3, b = a + 1$	$S(a, a + 2)$ with $\neg T(a, a + 2);$ $S(b, c);$ fixed $a + 2; T(b, a + 2)$	contradicted or routed by tree uniqueness and the right frontier
$c - a = 3, b = a + 2$	fixed $a + 1; S(a + 1, c); T(a + 1, b);$ adjacent $S(a + 1, b)$	absorbed by path uniqueness and adjacent-repeat normalization
$d - b = 3, c = b + 2$	$S(b, c), S(b + 1, c), S(b + 1, d),$ $T(b, b + 1), T(b, c), T(b + 1, c)$	absorbed by the right special-slice reductions

For example, in the slice

$$c - a = 3, \quad b = a + 1,$$

the remaining branch

$$a \not\sim_{\pi} a + 2, \quad a \sim_{S_{\pi}} a + 2$$

is contradictory. The repeat $a \sim_{S_{\pi}} a + 2$ copies one contour step into a path whose endpoints are already connected by the direct path coming from $a \sim_{\pi} c$. Tree uniqueness forces $a \sim_{\pi} a + 2$, contradiction.

In the other left gap-three slice,

$$c - a = 3, \quad b = a + 2,$$

the fixed branch at $a + 1$ and the branch $a + 1 \sim_{S_{\pi}} c$ are similarly absorbed by comparing two simple incidence paths. The checked right special slice

$$d - b = 3, \quad c = b + 2$$

is handled by the same principle, with left and right contour roles shifted.

7 The Conditional Bidefect Theorem

We now state the current reduction in journal-readable form. The hypotheses are exactly the remaining theorem-strength leaves.

Hypothesis 7.1 (Large-gap left adjacent links). For every left-tight π and every crossing quadruple $a < b < c < d$ with

$$a \sim_{\pi} c, \quad b \sim_{\pi} d, \quad a \not\sim_{\pi} b,$$

if

$$2 < c - a, \quad c - a \neq 3,$$

then the left adjacent link holds:

$$a \sim_{S_{\pi}} b.$$

Hypothesis 7.2 (Large-gap right adjacent links). Under the same crossing assumptions, if

$$2 < d - b$$

and the special checked slice

$$d - b = 3, \quad c = b + 2$$

does not occur, then the right adjacent link holds:

$$c \sim_{S_{\pi}} d.$$

Hypothesis 7.3 (Return-normalized nested skip collision). Let π be left-tight and assume its cycle partition is noncrossing. If x is a non-fixed predecessor skip and

$$x \not\sim_{\pi} \gamma(\pi x),$$

then the branch-normalized return interval supplied by the preceding lemma forces two distinct points y, z such that

$$y \sim_{\pi} z \quad \text{and} \quad y \sim_{S_{\pi}} z.$$

Hypothesis 7.4 (Chapuy slicing bound). The plus-defect classes admit the unicellular-map genus interpretation $\delta_+ = 2g$, and the corresponding one-sided genus slices satisfy

$$N_{m,g} \leq (2(m+1))^3 N_{m,g-1}$$

for all $g \geq 1$, with the genus-zero base bounded by 4^{m+1} .

Theorem 7.5 (Conditional bidefect Aubrun bound). *Assume the four hypotheses above. Then for every $m \geq 0$ and every*

$$0 \leq a \leq m + 3, \quad 0 \leq b \leq m + 1,$$

one has

$$N_m(a, b) \leq \mathcal{E}_m(a, b).$$

Proof. The rank budget and bidefect split reduce the analytic Wick sum to a two-parameter count over (a, b) . The plus-defect class controls every bidefect class with the same a , so it remains to prove the one-sided plus-defect bound.

The genus-zero plus-defect slice is the left-tight class. Noncrossing and predecessor reconstruction inject it into noncrossing partitions, giving the bound 4^{m+1} . The predecessor reconstruction follows by contradiction: a non-fixed predecessor skip first gives either the direct collision, already closed, or the nested collision supplied by the third hypothesis. In either case two distinct points lie in both a π -cycle and an S_{π} -cycle, contradicting left-tight transversality.

The finite gap-three Biane branches used in this reconstruction are accounted for in the table of Section 5 and handled by incidence-tree path-uniqueness reductions. The remaining large-gap adjacent links are exactly the first two hypotheses.

The positive plus-defect slices are then bounded by the Chapuy recurrence. Iterating from the genus-zero base gives the required polynomial envelope in the plus defect. Finally, the bidefect scalar absorption attaches the two binomial factors and pays the two defects by separate shifted bases. This is precisely the envelope $\mathcal{E}_m(a, b)$. $\square$

8 What Remains for an Unconditional Article

The theorem is a clean reduction, not yet the final Aubrun count theorem. To remove the word “conditional”, one must prove the four hypotheses in Section 6.

The first two are large-gap Biane adjacent-link theorems. The finite gap-three slices are no longer the issue; the remaining cases require a uniform argument for long contour gaps. The third is the nested skip collision theorem. The branch-normalized return interval lemma reduces its bookkeeping: every branch now provides a canonical long interval with a repeated contour class, and the current formal reduction makes precisely this return interval theorem-facing. The fourth is the Chapuy recurrence together with the exact conversion of plus-defect into unicellular genus.

A Synchronization with the Formal Development

This appendix is not part of the mathematical proof. It records the current source of the statements used above.

Bidefect split `wickPlusDefect_add_wickMinusDefect_eq_wickDefect`

Bidefect envelope `aubrunBiDefectCountEnvelope` and `AubrunBiDefectCountBound`

Plus-defect to bidefect count `wickBiDefectCount_le_biDefectEnvelope_of_plusDefectCount`

Genus-zero left-tight base `wickPlusGenusCount_zero_le_four_pow_of_no_nonfixed_cyclicPredecessorSkip`

Direct predecessor-skip collision `permCyclicPredecessorSkip_direct_commonCycle`

Branch-normalized nested interval `leftTight_nonfixed_nested_cyclicPredecessorSkip_exists_returnContourIntervalRepeat`

Current conditional endpoint `AubrunBiDefectCountBound_of_bianeAdjacentLinkHalves_left_gap_three_start_impossible_frontiers_nestedReturnContourIntervalCommonCycle_and_chapuy`

The current audit of the return-normalized endpoint reports only the standard foundational dependencies `propext`, `Classical.choice`, and `Quot.sound`. This is not a claim that U-AUB-02 is done; it is evidence that the nested collision leaf is now exposed through the checked branch-normalized return interval.

References

- [1] G. Aubrun, *Partial transposition of random states and non-centered semicircular distributions*, Random Matrices: Theory and Applications **1** (2012), no. 2, 1250001.
- [2] P. Biane, *Some properties of crossings and partitions*, Discrete Mathematics **175** (1997), no. 1–3, 41–53.
- [3] G. Chapuy, *A new combinatorial identity for unicellular maps, via a direct bijective approach*, Advances in Applied Mathematics **47** (2011), no. 4, 874–893.
- [4] J. Harer and D. Zagier, *The Euler characteristic of the moduli space of curves*, Inventiones Mathematicae **85** (1986), no. 3, 457–485.